① $Z_1 = \sum_{\{\mu\}} e^{-\beta \varepsilon_\mu}$ ortho $\Rightarrow E = \varepsilon$ triply degenerate
para $\Rightarrow E = 0$

a) $Z_1 = 3e^{-\beta\varepsilon} + e^{-\beta 0} = 1 + 3e^{-\beta\varepsilon}$

is the partition function for one molecule

for N molecules $Z = (1 + 3e^{-\beta\varepsilon})^N$ since they are non-interacting

$F = -kT \ln Z = -NkT \ln(1 + 3e^{-\beta\varepsilon})$

b) $n_o = N\langle n_o \rangle = N \dfrac{3e^{-\beta\varepsilon}}{1+3e^{-\beta\varepsilon}} = \dfrac{3N}{3+e^{\beta\varepsilon}}$ ortho

$n_p = N\langle n_p \rangle = \dfrac{N}{1+3e^{-\beta\varepsilon}}$ para

High-T: $n_o \to \frac{3}{4}N$, $n_p \to \frac{N}{4}$

equal distribution among energy states.
ortho is triplet $(\frac{3}{4}N)$ and para is singlet $(\frac{N}{4})$

Low-T: $n_o \to 0$ $n_p \to N$

All molecules in lowest energy state ($\varepsilon = 0$, para-hydrogen)

c) $E = -\dfrac{\partial}{\partial\beta} \ln Z = -N \dfrac{\partial}{\partial\beta} \ln(1 + 3e^{-\beta\varepsilon})$

$= -N \dfrac{1}{1+3e^{-\beta\varepsilon}} \cdot \left(-3\varepsilon e^{-\beta\varepsilon}\right) = \dfrac{3N\varepsilon}{3+e^{\beta\varepsilon}} = \dfrac{3N\varepsilon}{3+e^{\varepsilon/kT}}$

$C_V = \left(\dfrac{\partial E}{\partial T}\right)_V = \dfrac{3N\varepsilon(-1)}{(3+e^{\varepsilon/kT})^2}\left(\dfrac{-\varepsilon}{kT^2} e^{\varepsilon/kT}\right) = \dfrac{3N\varepsilon^2 e^{\varepsilon/kT}}{kT^2(3+e^{\varepsilon/kT})^2}$

At high T: $E \to \dfrac{3N\varepsilon}{4}$

Again see energy distribution according to probability

At low T: $E \to 0$

① cont...

For C_V:

At high-T: $C_V \to \frac{3N\varepsilon^2}{16kT^2}$

At low-T: $C_V \simeq \frac{3N\varepsilon^2 e^{\varepsilon/kT}}{kT^2(3+e^{\varepsilon/kT})^2} \sim \frac{3N\varepsilon^2 e^{\varepsilon/kT}}{kT^2 e^{2\varepsilon/kT}} \sim \frac{3N\varepsilon^2}{kT^2 e^{\varepsilon/kT}}$

so as $T \to 0$ $C_V \to 0$

② $\mathcal{H} = \frac{p_x^2+p_y^2}{2m}$

a) $Z_N = \frac{(Z_1)^N}{N!}$ for N indistinguishable particles

$$Z_1 = \int \frac{dx\,dy\,dp_x\,dp_y}{(2\pi\hbar)^2} e^{-\beta\left(\frac{p_x^2+p_y^2}{2m}\right)} = \frac{V}{(2\pi\hbar)^2}\left(\int_{-\infty}^{\infty} dp\, e^{-\frac{\beta p^2}{2m}}\right)^2$$

let $u = p\sqrt{\frac{\beta}{2m}}$ $\quad Z_1 = \frac{V}{(2\pi\hbar)^2}\left(\sqrt{\frac{2m}{\beta}}\int_{-\infty}^{\infty} du\, e^{-u^2}\right)^2$

$$Z_1 = \frac{V}{(2\pi\hbar)^2}\frac{2m}{\beta}\pi = \frac{L^2 m}{2\pi\hbar^2\beta}$$

$Z_N = \frac{(Z_1)^N}{N!}$ $\quad Z_N = \frac{L^{2N}}{N!}\left(\frac{2\pi m}{(2\pi\hbar)^2\beta}\right)^N$

b) $E = -\frac{\partial}{\partial\beta}\ln Z_N = -\frac{\partial}{\partial\beta} N\left(\ln\left(L^2 2\pi m\right) - \ln\left(N!^{1/N}(2\pi\hbar)^2\beta\right)\right)$

$E = \frac{N}{\beta} = NkT$

c) $S = -\left(\frac{\partial F}{\partial T}\right)_{N,V}$ $\quad F = -kT\ln Z_N = -kT\left(N\ln\left(\frac{L^2 2\pi mkT}{(2\pi\hbar)}\right) - \ln N!\right)$

$\ln N! = N\ln N - N$ $\quad F = -kT\left(N\ln\left(\frac{L^2 2\pi mkT}{N(2\pi\hbar)^2}\right) + N\right)$

$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N} = Nk\left(\ln\left(\frac{L^2 2\pi mkT}{N(2\pi\hbar)^2}\right)+1\right) + NkT\left(\frac{1}{T}\right)$

$$S = Nk\left(\ln\left(\frac{L^2 2\pi mkT}{N(2\pi\hbar)^2}\right)+2\right)$$

② cont...

d) $\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$ $\quad \mu = -kT\left(\ln\left(\frac{L^2 2\pi mkT}{N(2\pi\hbar)^2}\right)+1\right) - NkT\left(-\frac{1}{N}\right)$

$$\mu = -kT\left(\ln\left(\frac{L^2 2\pi mkT}{N(2\pi\hbar)^2}\right)\right)$$

③ $\varepsilon_n = (n+\frac{1}{2})\hbar\omega$

$$Z = \sum_n e^{-\varepsilon_n/kT} = \sum_{n=0}^{\infty} e^{-(n+\frac{1}{2})\hbar\omega/kT} = e^{-\frac{\hbar\omega}{2kT}}\sum_{n=0}^{\infty} e^{-\frac{\hbar\omega}{kT}n}$$

$$Z = \frac{e^{-\hbar\omega/2kT}}{1-e^{-\hbar\omega/kT}}$$

prob. to be in state k: $P_k = \frac{e^{-\varepsilon_k/kT}}{Z}$

prob. to be in odd-k state: $P_{k\,odd} = \sum_{k=odd} P_k$

odd k: $k = 2\ell+1$, $\ell = 0,1,2,\ldots$

$$P_{k\,odd} = \frac{1}{Z}\sum_{\ell=0}^{\infty} e^{-(2\ell+1+1/2)\hbar\omega/kT} = \frac{e^{-3\hbar\omega/kT}}{Z}\sum_{\ell=0}^{\infty} e^{-\frac{\hbar\omega}{kT}2\ell}$$

$$= \frac{e^{-3\hbar\omega/kT}}{Z}\left\{1+e^{-\frac{2\hbar\omega}{kT}}+\left(e^{-\frac{2\hbar\omega}{kT}}\right)^2+\ldots\right\}$$

$$= \frac{e^{-3\hbar\omega/kT}/Z}{1-e^{-2\hbar\omega/kT}} = \frac{e^{-3\hbar\omega/kT}}{1-e^{-2\hbar\omega/kT}}\cdot\frac{1-e^{-\hbar\omega/kT}}{e^{-\hbar\omega/kT}}$$

$$= \frac{e^{-\hbar\omega/kT}}{1+e^{-\hbar\omega/kT}} = \frac{1}{e^{\hbar\omega/kT}+1}$$

limits $T\to\infty$ $\quad P_{k\,odd} = \frac{1}{2}$ equal probability to be in even or odd state

$T\to 0$ $\quad P_{k\,odd} = e^{-\hbar\omega/kT}$ S.H.O in ground state $n=0$ which is even

4 a) $Z_N = \left(\frac{Z_a^{N_a}}{N_a!}\right)\left(\frac{Z_b^{N_b}}{N_b!}\right)$ if $Z_a = Z_b$ $\quad Z_N = \frac{Z_a^{N_a+N_b}}{N_a!\,N_b!} = \frac{Z_a^{N}}{N_a!\,N_b!}$

$$F = -kT\ln Z = -kTN\ln Z_a + kT\ln(N_a!\,N_b!)$$

$$\tilde{F} = -kT\ln\tilde{Z}_N \qquad \tilde{Z}_N = \frac{Z_a^N}{N!}$$

$$\tilde{F} = -kTN\ln Z_a + kT\ln N!$$

b) $S = -\frac{\partial F}{\partial T} = Nk\ln Z_a + NkT\frac{\partial}{\partial T}\ln Z_a - k\ln(N_a!\,N_b!)$

$$\tilde{S} = -\frac{\partial \tilde{F}}{\partial T} = Nk\ln Z_a + NkT\frac{\partial}{\partial T}\ln Z_a - k\ln N!$$

$$\delta S = S - \tilde{S} = k\ln N! - k\ln N_a!\,N_b! = k\ln\frac{N!}{N_a!\,N_b!}$$

$$\frac{\delta S}{k} = N\ln N - N - (N_a\ln N_a - N_a + N_b\ln N_b - N_b)$$

$$\frac{\delta S}{k} = N\ln N - N_a\ln N_a - N_b\ln N_b$$

$$\frac{\delta S}{k} = N_a\ln\frac{N}{N_a} + N_b\ln\frac{N}{N_b}$$